

DEEP DIVE

Backend Engineering

APIs, databases, caching and production systems — 10 weeks

 10 weeks

 Intermediate

 Live+Recorded

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ABOUT THIS COURSE

A thorough 10-week backend engineering program. Build production-grade APIs, master database design, caching, auth, and deployment — with real projects every two weeks.

Prerequisites: Python basics. Basic understanding of APIs (HTTP, JSON).

WHAT YOU'LL LEARN

- 1 Build and ship production REST APIs
- 2 Design and optimise relational databases
- 3 Implement auth systems securely
- 4 Deploy backend services to the cloud

CURRICULUM · 5 SESSIONS

SESSION WEEKS 1–2

FastAPI & REST

REST API design principles FastAPI in depth — routing, dependencies, middleware
Request validation with Pydantic Project 1: production-ready REST API

SESSION WEEKS 3–4

Databases

PostgreSQL in depth — schemas, indexes, transactions ORM with SQLAlchemy / Prisma
Database design & normalisation Query optimisation

SESSION WEEKS 5–6

Auth & Security

JWT authentication & refresh tokens

OAuth 2.0 flows

Role-based access control (RBAC)

OWASP top 10 for APIs

SESSION WEEKS 7–8

Caching & Async

Redis — caching, sessions, pub-sub

Background tasks & queues (Celery / BullMQ)

WebSockets & real-time

Project 2: real-time chat API

SESSION WEEKS 9–10

Deployment & Production

Docker & docker-compose

CI/CD with GitHub Actions

Cloud deployment (AWS / Vercel)

Capstone: full backend system

WHAT'S INCLUDED

✓ 30 live sessions + all recordings

✓ Architecture reference guide

✓ All project source code

✓ Learner's Certificate

UPCOMING BATCHES



Ongoing · Mon, Wed, Fri · 8:00 PM – 9:30 PM IST



New batches start monthly



learner

₹16,999

Live+Recorded · 10 weeks

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